Mithlesh Upadhyay

Software Engineer • Backend & Full-Stack • Python, TypeScript, AI

Delhi, India | mithleshupadhyay515@gmail.com
Portfolio | **GitHub** | **LinkedIn** | **Wellfound**

CONTACT

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CORE SKILLS

Languages: Python, TypeScript, JavaScript, SQL, PHP, Bash; working knowledge of Go, Rust

Backend: FastAPI, AsyncIO, Pydantic, SQLAlchemy, Alembic, REST APIs, WebSockets, Microservices

Frontend: React, TypeScript, Vite, TanStack Query, Tailwind CSS, Next.js, Astro, WordPress

Data & Messaging: PostgreSQL, Redis, Kafka, Celery, MongoDB, OpenSearch, MinIO/S3

AI & LLM: LangGraph, LangChain, LiteLLM, OpenAI, Gemini, Anthropic, RAG, Agents, pgvector, FAISS

DevOps & Cloud: Docker, Docker Compose, GitHub Actions, Caddy, Nginx, DigitalOcean, Linux, Prometheus

Testing & Quality: pytest, Playwright, Vitest, Testing Library, Nox, Ruff, mypy

Security & Auth: JWT/OIDC, OAuth, RBAC, Multi-tenancy, Rate limiting, PII masking, Secret scanning

Integrations: Razorpay, Cashfree, PayU, UPI, WhatsApp Cloud API, SMTP, SMS OTP

EDUCATION

M.Tech, Artificial Intelligence
Delhi Technological University
2021 – 2023

B.E., Computer Science
Vindhya Institute of Technology & Science
2008 – 2012

PROFESSIONAL SUMMARY

- Software engineer with **4+ years** of experience shipping production backends, web apps, and AI features in Python and TypeScript.
- Sole full-stack engineer of a live IoT ad marketplace: designed, built, deployed, and now maintain its FastAPI backend, React apps, edge runtime, and infrastructure.
- Strong in FastAPI, PostgreSQL, Redis, Kafka, Celery, Docker, and CI/CD, with React, TypeScript, and Vite on the frontend.
- Worked end to end across an AI security platform: LLM integrations, RAG and agent workflows, red teaming, and test automation.

PROFESSIONAL EXPERIENCE

Senior Software Engineer (Contract) | Radoflex

Aug 2026 – Present | Remote, India

myRado: hyperlocal digital out-of-home (DOOH) ad network of in-store LED screens and RFID-verified print ads. Live: myrado.io · app.myrado.io · admin.myrado.io

- Took myRado from first commit to production in **6 weeks** as its sole full-stack engineer, then owned fixes, releases, and maintenance across 8 repositories and 800+ commits.
- Architected a FastAPI + PostgreSQL modular monolith (Python 3.13, async SQLAlchemy, 83 Alembic migrations) with **270+ REST endpoints** across 7 domains.
- Integrated Razorpay, Cashfree, PayU, and UPI with EMI collections, GST invoicing, and partner payouts; added phone OTP, Google sign-in, and JWT RBAC.
- Built an FFmpeg media worker on a Postgres-backed job queue (leases, retries, backoff), MinIO presigned uploads, and WhatsApp, email, and SMS alerts.
- Built the Python edge runtime and WebSocket device gateway (auth, heartbeats, offline queueing, signed OTA) and integrated the embedded engineer's LED controller SDK.
- Shipped a 25-screen React 19 + TypeScript admin console, a mobile-first advertiser app (Vite, TanStack Query, OpenAPI-typed client), and an Astro site.
- Deployed on DigitalOcean with Docker Compose and Caddy auto-TLS across 5 subdomains, plus nightly Postgres and MinIO backups and a restore runbook.
- Backed delivery with **2,400+ automated tests** (pytest, Vitest), CI quality gates (Ruff, mypy, Alembic drift checks, secret scans), and 110+ QA scenarios.
- Owned product and engineering docs (requirements, architecture, 9 ADRs, QA plans, runbook), working directly with business stakeholders and an embedded engineer.

Founder & Full-Stack Developer | w3colleges.org

May 2026 – Present (part-time since Aug 2026) | Delhi, India

Learning platform: 443 CS tutorials, GATE exam prep, courses, and engineering services.

- Rebuilt the site end to end as a custom WordPress/PHP theme of **48 modules** (260+ commits), owning backend, frontend, and all third-party integrations.
- Built a paid GATE CS mock-test system: timed exam engine, Razorpay checkout, coupons, invoices, result analytics, reminders, and REST APIs.
- Added quizzes, coding practice with a browser IDE and code judge (Wandbox/Piston), leaderboards, and Google, GitHub, and LinkedIn sign-in.
- Set up GitHub Actions CI/CD (PHP lint, rsync-over-SSH deploys, backups, smoke tests) plus SEO schema, caching, transactional email, and AdSense.

PUBLICATION

A Comprehensive Analysis of Sentiment Analysis [Paper]
IEEE, 2024

SELECTED PROJECTS

Agentic RAG [Repo] [Demo]
Multi-tenant RAG with an 8-node LangGraph agent, hybrid BM25 + pgvector retrieval, grounding checks, Kafka ingestion, RBAC, and 350+ tests.

Multi-Agent Orchestration [Repo]
Real-time multi-agent orchestration with shared context, tool-failure contracts, context budgets, multi-hop retrieval, and evals.

Skyclad LLM Gateway [Repo]
Multi-tenant LLM gateway in TypeScript: unified chat API, streaming, provider routing, budgets, and caching across OpenAI, Anthropic, and Gemini.

Payment Processing Service [Repo]
Payment initiation and status tracking with idempotency keys, retries, concurrency control, gateway webhooks, and traceable events.

ChadWallet Web [Repo] [Live]
Next.js landing and trading page on Solana APIs, with Privy sign-in (Google and Apple).

AI Backend [Repo]
Asynchronous AI Q&A backend with FastAPI, Celery, Redis, OpenAI, tests, and a Dockerized setup.

Software Engineer II | Emulate AI (formerly Detoxio AI)

Dec 2024 – Apr 2026 | Remote, India

AI security startup: LLM red teaming, AI-assisted pentesting, and AI data protection.

- Worked end to end on CoPen, an AI pentesting platform: FastAPI services, Celery workers, a React 19 app, and LiteLLM-based analysis on a 20-service Docker stack.
- Built LLM red-teaming features in the dtx CLI/SDK and assessment service: jailbreak prompt transforms, LiteLLM provider targets, and risk-report APIs.
- Built Prompt Guard for LLM I/O safety: PII and secret detection with Presidio, NER, and an LLM detector, plus reversible, format-preserving masking via an SDK.
- Delivered a GenAI SOC assistant for a client: LangChain ReAct and Text-to-SQL agents, Kafka-fed false-positive reduction, and 50K-event novelty detection.
- Built multi-tenant FastAPI services for the next-generation platform (25+ data models, auth, campaign and recon APIs) and an event and object store with Kafka streaming.
- Built pytest unit, integration, and E2E suites across the SDK, plugins, scanners, proxies, and browser agents; reached **80% coverage** and a Nox matrix on Python 3.11–3.14.
- Ran the stack in Docker Compose and CI with GitHub Actions PR validation; fixed build timeouts, submodule pipelines, and service-initialization issues.

SDE-I (AI Engineer) | Propellyr Inc.

Jan 2023 – Aug 2024 | Delhi, India

- Built and maintained FastAPI services and REST APIs for AI chatbot, retrieval, and model-serving workflows using async request handling and modular service layers.
- Developed RAG pipelines and chatbot workflows using LangChain, document chunking, embeddings, FAISS retrieval, context assembly, and prompt orchestration.
- Fine-tuned LLaMA 2 and other open-source LLMs with QLoRA on proprietary datasets, from training-data preparation to inference-quality evaluation.
- Implemented distributed background processing with Celery and Redis for ingestion, embedding generation, model tasks, retries, and status tracking.
- Integrated Kafka-based event flows to decouple backend services and coordinate asynchronous processing across service boundaries.
- Containerized services with Docker and Docker Compose and added Pytest unit and integration coverage for APIs, service logic, database workflows, and workers.

Python Developer | Foresiet

Sep 2022 – Dec 2022 | Remote, India

- Built Python crawler utilities for automated APK data extraction, metadata processing, and cybersecurity research workflows.
- Developed FastAPI services and REST endpoints exposing extraction workflows, metadata results, and automation capabilities to downstream tools.
- Created Node.js integration utilities for API orchestration and file processing, plus React UI components for reviewing crawl status and results.

Senior Technical Content Engineer | GeeksforGeeks

Sep 2017 – Dec 2021 | Noida, India

- Led technical content and course development for 4+ years across computer science and competitive-exam domains, including GATE CS.
- Published and refined large-scale technical content: **7,000+ articles**, 3,000+ improvements, and 9,000+ review contributions.
- Managed and mentored a team of contributors while maintaining quality, consistency, and delivery timelines across multiple learning products.